

Bony fish

About 400 million years ago, a new family of fish began swimming in the seas. Unlike the sharks that had ruled the waters for millions of years, the new fish had skeletons hardened with calcium to form bone, earning them the name “bony fish.” The bony fish evolved into a huge range of new species and make up more than 95% of fish species alive today.

Leedsichthys

LEEDS-ick-thiss

- **When** 176–161 million years ago (Middle Jurassic)
- **Fossil location** Europe, Chile
- **Habitat** Oceans
- **Length** 30 ft (9 m)

Perhaps the largest bony fish that ever lived, *Leedsichthys* was bigger than a killer whale. Despite its fearsome size, it was a harmless filter feeder rather than a hunter—it

gulped huge volumes of water into its mouth and then squirted it out while sifting shrimp and other tiny animals with its gills. Bite marks on one fossil show that *Leedsichthys* was hunted by gigantic marine reptiles called pliosaurs.



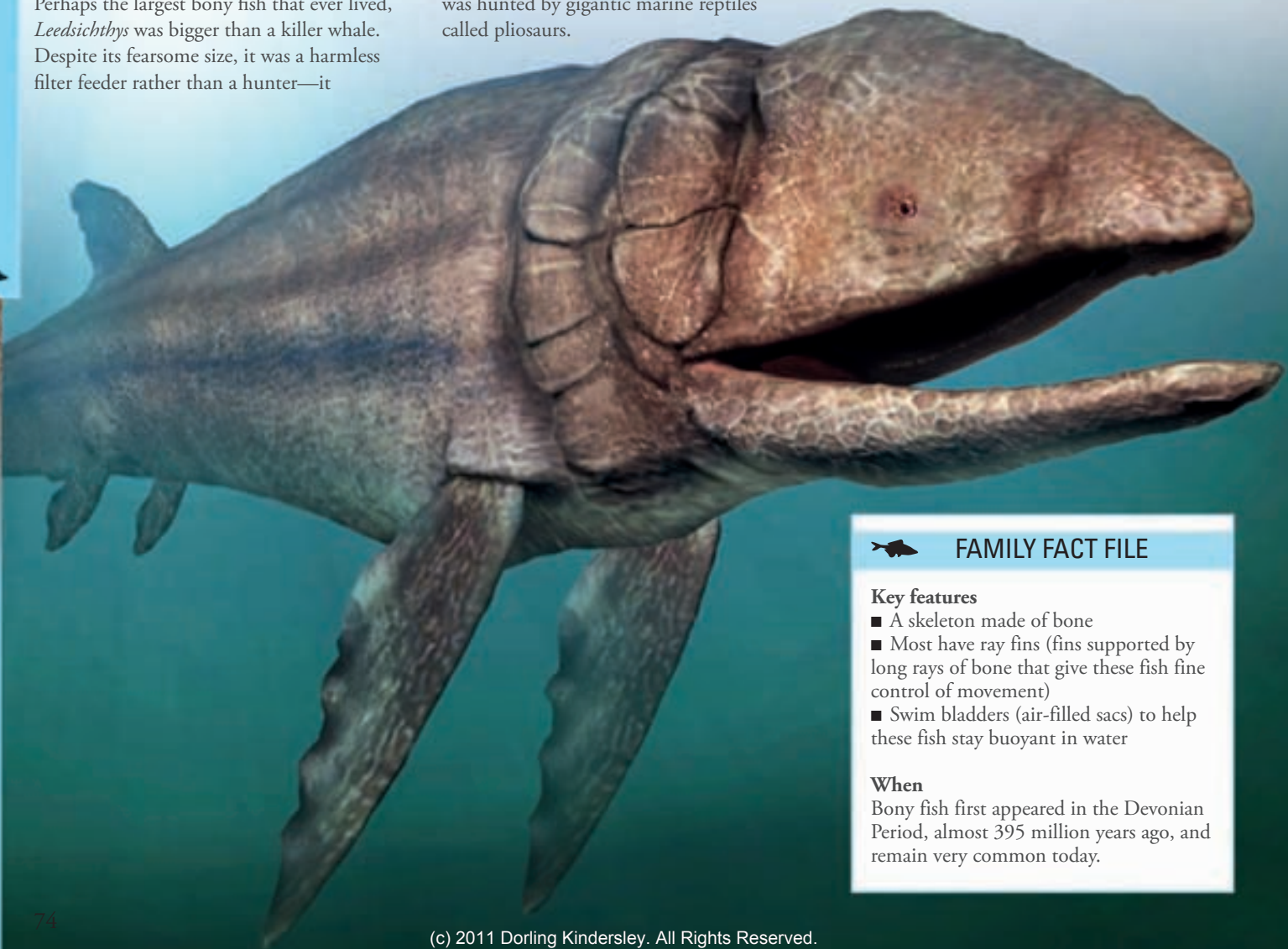
Xiphactinus

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- **When** 112–70 million years ago (Middle to Late Cretaceous)
- **Fossil location** N. America
- **Habitat** Shallow waters of N. America
- **Length** 20 ft (6 m)

Xiphactinus was a powerful swimmer with a long, muscular body. It had a huge mouth and could swallow large prey whole. One fossil was found to contain the remains of a 7 ft (2 m) fish in its stomach—perhaps the prey was too big for *Xiphactinus* and killed it by thrashing around inside.



FAMILY FACT FILE

Key features

- A skeleton made of bone
- Most have ray fins (fins supported by long rays of bone that give these fish fine control of movement)
- Swim bladders (air-filled sacs) to help these fish stay buoyant in water

When

Bony fish first appeared in the Devonian Period, almost 395 million years ago, and remain very common today.